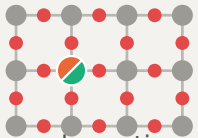
Pr and Rb both lower the O-vacancy cost in anatase TiO_2; only Rb makes it spontaneous

Spin-polarised PBE+U, U(Ti-3d) = 3.5 eV · oxygen-rich reference $\mu_O = \frac{1}{2}E(O_2)$ · one dopant per 48-atom cell

❶ Build cell



Ti O Pr/Rb

2×2×1 anatase, 48 atoms

One Ti → Pr^{3+} or Rb^{+} (separate cells)

❷ Remove O



First-shell O of the dopant

V_O releases

●● 2 e^-

❸ Protocol

Γ-only relaxations, 40 Ry

$E_f = E(V_O) - E(\text{perfect}) + \frac{1}{2}E(O_2)$

Pr–Rb contrast, 5.36 eV

✓ 2×2×2 k-mesh, 55/70 Ry, D3: shifts ≤ 0.04 eV

! smearing, U(O-2p): Pr spin state changes

narrower smearing: Pr +0.99 eV, contrast 5.11 eV

❹ Charge bookkeeping

○ hole h^+ ● electron e^-

● **Pr^{3+}**

dopant V_O net

○ + ●● → ● **1 e^- left**

extra e^- in Ti-3d conduction band

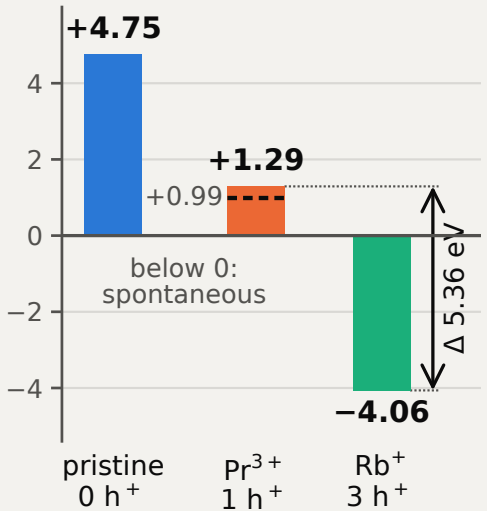
● **Rb^+**

○○○ + ●● → ○ **1 h^+ left**

residual h^+ on O-2p: 0.59 μ_B on one O bonded to Rb

Rb shifts 1.27 Å towards V_O

❺ Result: $E_f(V_O)$ (eV)



Dashed: Pr, narrower smearing